Exact zCDP for k-Ary Randomized Response

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Abstract

We determine the least zero-concentrated differential privacy parameter of k-ary randomized response for every $k \geq 2$ and every privacy parameter $\varepsilon \geq 0$. The necessary condition for endpoint optimality obtained by Harrison and Manurangsi is also sufficient, including the equality case. When this condition fails, the normalized Rényi-divergence objective has a unique maximizer, characterized by a scalar equation with an explicit finite bracket. The proof expresses the objective as a secant slope of a function whose curvature changes sign at most once. This characterization also yields bisection with geometric error bounds for both the maximizing order and the optimal zCDP parameter, in exact real arithmetic.

1 Introduction and preliminaries

Zero-concentrated differential privacy (zCDP), introduced by Bun and Steinke [2], bounds Rényi divergence simultaneously at every order $\alpha > 1$. Determining the least admissible parameter therefore requires a global optimization over the order. Harrison and Manurangsi determine this parameter for several fundamental mechanisms. For k-ary randomized response, they establish the endpoint formula for $2 \leq k \leq 6$ and give a necessary condition for this formula to hold for general k [4, Theorem 16 and Lemma 17]. They also note that their method extends to moderate values of $k > 6$ under more complicated constraints [4, footnote 4], while leaving a precise characterization for large k as an open problem [4, Section 7]. We prove that their threshold is exact and characterize the optimum above it by a unique root of an explicit scalar equation.

Fix an integer $k \geq 2$ and write $[k] = \{1, \ldots, k\}$. For $\varepsilon \geq 0$, the mechanism $\mathrm{RR}_{k,\varepsilon}$ takes values in $[k]$ and has output law P_i on input i, where

$$P_i(j) = \Pr[\mathrm{RR}_{k,\varepsilon}(i) = j] = \begin{cases} e^{\varepsilon}/(e^{\varepsilon} + k - 1), & j = i, \\ 1/(e^{\varepsilon} + k - 1), & j \neq i. \end{cases}$$

Every pair of distinct inputs is adjacent. For positive probability mass functions P, Q on $[k]$ and $\alpha > 1$, the Rényi divergence is

$$\mathrm{D}_{\alpha}(P\|Q) = \frac{1}{\alpha - 1} \log \sum_{j=1}^{k} P(j)^{\alpha} Q(j)^{1-\alpha}.$$

The mechanism is ρ-zCDP, for $\rho \geq 0$, if

$$\mathrm{D}_{\alpha}(P_i\|P_j) \leq \rho\alpha \qquad (i, j \in [k],\ \alpha > 1).$$

Let $\rho_*(k, \varepsilon)$ denote the least such ρ, and set $q(\alpha) = \mathrm{D}_{\alpha}(P_1\|P_2)/\alpha$. All logarithms are natural.

For $Y \sim P_1$, define the privacy loss $L = \log(P_1(Y)/P_2(Y))$ and its cumulant generating function $\Lambda(s) = \log \mathbb{E}[e^{sL}]$. The variable L has at most three mass points. Atouani, Marchal, and Arbel [1] study optimal sub-Gaussian variance proxies for three-mass distributions. For centered privacy loss, this proxy is $\sup_{s\neq 0} 2[\Lambda(s) - s\Lambda'(0)]/s^2$, whereas the least zCDP parameter is $\sup_{s>0} \Lambda(s)/[s(s+1)]$. The latter leads to a secant-slope problem with a positive base point.

2 The exact parameter

Theorem 1. *For every integer $k \geq 2$, $\rho_*(k, 0) = 0$. For $\varepsilon > 0$, put*

$$a = \frac{\varepsilon}{2}, \qquad c = \frac{k-2}{2e^a}, \qquad K(\varepsilon) = \frac{2(e^\varepsilon - 1)(e^\varepsilon - 1 - \varepsilon)}{\varepsilon(e^\varepsilon + 1) - 2e^\varepsilon + 2}.$$

The denominator defining $K(\varepsilon)$ is strictly positive, and the following alternatives hold.

1. *If $k \leq K(\varepsilon)$, including equality, then*

$$\rho_*(k, \varepsilon) = \frac{\varepsilon(e^\varepsilon - 1)}{e^\varepsilon + k - 1}. \tag{1}$$

   The function q is strictly decreasing on $(1, \infty)$, and its supremum is its limit as $\alpha \downarrow 1$.

2. *If $k > K(\varepsilon)$, the equation*

$$(x^2 - a^2)\sinh x = 2x(c + \cosh x)\log\frac{c + \cosh x}{c + \cosh a} \tag{2}$$

   has exactly one solution x_* in the domain $x > a$. In this case

$$\rho_*(k, \varepsilon) = \frac{\varepsilon^2 \sinh x_*}{2x_*(c + \cosh x_*)}, \qquad \alpha_* = \frac{1}{2} + \frac{x_*}{\varepsilon} \tag{3}$$

   and $\alpha_* > 1$ is the unique maximizer of q. Moreover,

$$a < x_* < U := 2\log\big(2(c + \cosh a)\big) = 2\log(k - 1 + e^\varepsilon) - \varepsilon. \tag{4}$$

For every $\varepsilon \geq 0$ and $\rho \geq 0$, the mechanism $\mathrm{RR}_{k,\varepsilon}$ is ρ-zCDP if and only if $\rho \geq \rho_*(k, \varepsilon)$.

The threshold K is due to Harrison and Manurangsi [4, Lemma 17]. The endpoint $x = a$ also satisfies (2), but lies outside the domain $x > a$.

3 Proof of the characterization

If $\varepsilon = 0$, every output law is uniform, so all the divergences vanish. Henceforth assume $\varepsilon > 0$.

For any distinct inputs i, j, only the probabilities of symbols i and j differ. Substituting their values gives

$$\mathrm{D}_\alpha(P_i \| P_j) = \frac{1}{\alpha - 1}\log\frac{e^{\alpha\varepsilon} + e^{(1-\alpha)\varepsilon} + k - 2}{e^\varepsilon + k - 1}. \tag{5}$$

This is the expression in [4, Proposition 20]. It is independent of the ordered pair of distinct inputs; equal inputs have divergence zero. Consequently

$$\rho_* = \sup_{\alpha > 1} q(\alpha). \tag{6}$$

Set

$$x = \varepsilon\left(\alpha - \frac{1}{2}\right) > a, \qquad h(x) = \log(c + \cosh x), \qquad t = x^2, \quad t_0 = a^2,$$

and define

$$f(t) = h(\sqrt{t}) \quad (t > 0), \qquad S(t) = \frac{f(t) - f(t_0)}{t - t_0} \quad (t > t_0).$$

The numerator and denominator inside the logarithm in (5) are $2e^a(c + \cosh x)$ and $2e^a(c + \cosh a)$, respectively. Since $\alpha(\alpha - 1) = (x^2 - a^2)/\varepsilon^2$, we obtain

$$q(\alpha) = \varepsilon^2 S(t), \qquad \rho_* = \varepsilon^2 \sup_{t > t_0} S(t). \tag{7}$$

The curvature pattern below is equivalent, under $u = \sinh x$, to the monotonicity result of Guo, Luo, and Qi [3, Theorem 1].

Lemma 1 (Curvature)**.** *For $c \geq 0$, define*

$$g(x) = x(1 + c\cosh x) - \sinh x(c + \cosh x), \qquad x \geq 0.$$

For $x > 0$,

$$f'(x^2) = \frac{\sinh x}{2x(c + \cosh x)}, \qquad f''(x^2) = \frac{g(x)}{4x^3(c + \cosh x)^2}. \tag{8}$$

If $c \leq 2$, then $g(x) < 0$ for all $x > 0$. If $c > 2$, there is a unique $b > 0$ such that $g(b) = 0$, with $g > 0$ on $(0, b)$ and $g < 0$ on (b, ∞).

Proof. Differentiation gives (8) and

$$g'(x) = \sinh x\big(cx - 2\sinh x\big). \tag{9}$$

The function $\sinh x/x$ increases strictly from 1 to infinity on $(0, \infty)$: its derivative has the sign of $x\cosh x - \sinh x$, which vanishes at zero and has derivative $x\sinh x > 0$.

For $c \leq 2$, equation (9) gives $g' < 0$ on $(0, \infty)$; as $g(0) = 0$, this proves the first case. For $c > 2$, there is a unique $r > 0$ with $2\sinh r/r = c$. Thus g increases strictly on $(0, r)$ and decreases strictly on (r, ∞), with $g(r) > 0$. Finally,

$$\lim_{x \to \infty} e^{-2x} g(x) = -\frac{1}{4}.$$

It follows that g has exactly one positive zero $b > r$, with the stated signs. □

The interior-maximum argument below is an instance of Yang's l'Hospital-type monotonicity rule [5, Theorem 8]; the direct proof also provides the explicit bracket U.

Lemma 2 (Secant slopes)**.** *If $g(a) \leq 0$, then S is strictly decreasing on (t_0, ∞). If $g(a) > 0$, then S increases strictly up to a unique $t_* > t_0$ and decreases strictly thereafter. In the latter case $c > 2$, $a < b < \sqrt{t_*} < U$, and t_* is the unique zero in (t_0, ∞) of*

$$J(t) = (t - t_0)f'(t) - f(t) + f(t_0). \tag{10}$$

Proof. The function J extends to t_0 with $J(t_0) = 0$, and

$$S'(t) = \frac{J(t)}{(t - t_0)^2}, \qquad J'(t) = (t - t_0)f''(t).$$

If $g(a) \le 0$, Lemma 1 implies $g(x) < 0$ for every $x > a$, including when $g(a) = 0$. Hence $J' < 0$ on (t_0, ∞), so $J < 0$ and $S' < 0$ there.

If $g(a) > 0$, the same lemma gives $c > 2$ and $a < b$. The function J increases strictly on (t_0, b^2) and decreases strictly on (b^2, ∞); in particular, $J(b^2) > 0$. Put $C = \log(2(c + \cosh a))$, so $U = 2C$. Since $2(c + \cosh a) > e^a$, we have $C > a > 0$. For $x \ge a$, the inequalities $h'(x) \le 1$ and $h(x) \ge x - \log 2$ give

$$J(x^2) = \frac{x^2 - a^2}{2x} h'(x) - h(x) + h(a) \le -\frac{x}{2} - \frac{a^2}{2x} + C.$$

Thus $J(U^2) \le -a^2/(4C) < 0$. The monotonicity of J now gives exactly one zero $t_* \in (b^2, U^2)$, with $J > 0$ on (t_0, t_*) and $J < 0$ on (t_*, ∞). The formula for S' proves the assertion. □

Proof of Theorem 1. To identify the threshold, write

$$A = a \cosh a - \sinh a, \qquad B = \sinh a \cosh a - a.$$

Both are strictly positive for $a > 0$, since they vanish at zero and have derivatives $a \sinh a$ and $2 \sinh^2 a$, respectively. As $g(a) = cA - B$,

$$\begin{aligned} g(a) \le 0 &\iff c \le B/A \\ &\iff k \le 2 + 2e^a \frac{\sinh a \cosh a - a}{a \cosh a - \sinh a} \\ &\iff k \le \frac{2(e^{2a} - 1)(e^{2a} - 1 - 2a)}{2a(e^{2a} + 1) - 2e^{2a} + 2} = K(\varepsilon). \end{aligned}$$

The denominator is $4e^a A > 0$. Thus this equivalence also includes equality.

If $k \le K(\varepsilon)$, Lemma 2 and (7) give

$$\rho_* = \varepsilon^2 f'(a^2) = \frac{\varepsilon^2 \sinh a}{2a(c + \cosh a)} = \frac{\varepsilon(e^\varepsilon - 1)}{e^\varepsilon + k - 1}.$$

Both changes of variable are strictly increasing, so q is strictly decreasing and its supremum is approached as $\alpha \downarrow 1$.

If $k > K(\varepsilon)$, let $x_* = \sqrt{t_*}$, where t_* is supplied by Lemma 2. Multiplying $J(t_*) = 0$ by $2x_*(c + \cosh x_*)$ gives (2); the lemma proves uniqueness on $x > a$ and the bracket (4). Moreover,

$$\sup_{t > t_0} S(t) = S(t_*) = f'(t_*),$$

which proves (3) and uniqueness of the maximizing order.

Finally, equation (6) shows that every $\rho \ge \rho_*$ satisfies all the zCDP inequalities. If $\rho < \rho_*$, an inequality for distinct inputs fails at α_* in the interior regime, or at an order sufficiently close to 1 in the endpoint regime. Together with the $\varepsilon = 0$ case, this proves the stated equivalence. □

4 Computation

Throughout this section, assume $\varepsilon > 0$ and $k > K(\varepsilon)$. Define, for $x \ge a$,

$$F(x) = \frac{x^2 - a^2}{2x} \frac{\sinh x}{c + \cosh x} - \log \frac{c + \cosh x}{c + \cosh a} = J(x^2). \tag{11}$$

Lemma 2 gives

$$F(x) > 0 \quad (a < x < x_*), \qquad F(x_*) = 0, \qquad F(x) < 0 \quad (x > x_*). \tag{12}$$

This sign pattern permits bisection from (a, U), despite the additional zero $F(a) = 0$ at the excluded endpoint.

Corollary 1. *Assume $\varepsilon > 0$ and $k > K(\varepsilon)$. Set $L_0 = a$, $R_0 = U$, and $m_n = (L_n + R_n)/2$. If $F(m_n) > 0$, set $(L_{n+1}, R_{n+1}) = (m_n, R_n)$; if $F(m_n) < 0$, set $(L_{n+1}, R_{n+1}) = (L_n, m_n)$; if $F(m_n) = 0$, terminate with $x_* = m_n$. Before exact termination,*

$$L_n < x_* < R_n, \qquad R_n - L_n = (U - a)2^{-n}.$$

The corresponding enclosure for α_* is $\left(\frac{1}{2} + L_n/\varepsilon, \frac{1}{2} + R_n/\varepsilon\right)$. The quantities

$$\ell_n = \frac{\varepsilon^2 h'(L_n)}{2R_n}, \qquad u_n = \frac{\varepsilon^2 h'(R_n)}{2L_n} \tag{13}$$

satisfy $\ell_n \le \rho_* \le u_n$, converge to ρ_*, and obey

$$u_n - \ell_n \le (\varepsilon + 2)(U - a)2^{-n}. \tag{14}$$

Thus an enclosure of width at most $\delta > 0$ is obtained within

$$\max\left\{0, \left\lceil \log_2 \frac{(\varepsilon + 2)(U - a)}{\delta} \right\rceil\right\}$$

bisection steps, or the exact root is found earlier.

Proof. The bracket invariant follows from (12). Differentiation gives

$$h''(x) = \frac{1 + c\cosh x}{(c + \cosh x)^2} > 0.$$

Thus h' is increasing, and $L_n \le x_* \le R_n$ together with (3) gives $\ell_n \le \rho_* \le u_n$. The shrinking bracket and continuity give convergence.

For $x > 0$, we have $0 \le h'(x) \le 1$ and $h''(x) \le 1$, the latter following from

$$(c + \cosh x)^2 - (1 + c\cosh x) = c^2 + c\cosh x + \sinh^2 x \ge 0.$$

Writing $L = L_n$, $R = R_n$, and $w = R - L$, we therefore have

$$\begin{aligned} u_n - \ell_n &= \frac{\varepsilon^2}{2}\left(\frac{h'(R) - h'(L)}{L} + h'(L)\left(\frac{1}{L} - \frac{1}{R}\right)\right) \\ &\le \frac{\varepsilon^2}{2}\left(\frac{w}{a} + \frac{w}{a^2}\right) = (\varepsilon + 2)w. \end{aligned}$$

Substituting $w = (U - a)2^{-n}$ proves (14) and the iteration bound. □

Corollary 1 assumes exact evaluation of elementary real functions and exact sign comparisons; its iteration bound is not a bit-complexity bound. A rigorous numerical implementation must also control rounding in the threshold comparison, the sign tests, and the value enclosure.

AI use disclosure

The author used AI tools (GPT-6 Astra Max and Opus 5.5 Max) for code review, assistance with manuscript translation, and the development of counterexamples. The author takes full responsibility for the code, the original manuscript, the final translation, and all other related outputs.

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